www.Padasalai.Net www.TrbTnpsc.com

COMMON QUARTERLY EXAMINATION - 2026

Standard - 12

Reg.No. | | | | | | |

MATHEMATICS

Time Allowed: 3.00 Hours *Maximum Marks: 90*

Part-A

Choose the correct answer: **20×1=20**

1) If A is a 3 × 3 non-singular matrix such that $AA^T = A^TA$ and $B = A^{-1}A^T$, then $BB^T =$
a) A b) B c) I_3 d) B^T

2) The value of $\sum_{n=1}^{13}(i^n + i^{n-1})$ is
a) 1+i b) i c) 1 d) 0

3) The number of positive zeros of the polynomial $\sum_{r=0}^{n} {}^nC_r(-1)^r x^r$ is
a) 0 b) n c) < n d) r

4) If $\vec{a}+\vec{b}+\vec{c}=0, |\vec{a}|=3, |\vec{b}|=4, |\vec{c}|=5$ then the angle between $\vec{a}$ and $\vec{b}$ is
a) $\frac{\pi}{6}$ b) $\frac{2\pi}{3}$ c) $\frac{5\pi}{3}$ d) $\frac{\pi}{2}$

5) If $|z_1|=1$, $|z_2|=2$, $|z_3|=3$ and $|9z_1z_2 + 4z_1z_3 + z_2z_3| = 12$, then the value of $|z_1 + z_2 + z_3|$ is
a) 1 b) 2 c) 3 d) 4

6) If $\sin^{-1}x = 2\sin^{-1}\alpha$ has a solution, then
a) $|\alpha| \le \frac{1}{\sqrt{2}}$ b) $|\alpha| \ge \frac{1}{\sqrt{2}}$ c) $|\alpha| < \frac{1}{\sqrt{2}}$ d) $|\alpha| > \frac{1}{\sqrt{2}}$

7) If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$; then $\cos^{-1}x + \cos^{-1}y$ is equal to
a) $\frac{2\pi}{3}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) π

8) The distance between the planes $x + 2y + 3z + 7 = 0$ and $2x + 4y + 6z + 7 = 0$
a) $\frac{\sqrt{7}}{2\sqrt{2}}$ b) $\frac{7}{2}$ c) $\frac{\sqrt{7}}{2}$ d) $\frac{7}{2\sqrt{2}}$

9) The domain of the function defined by $f(x) = \sin^{-1}\sqrt{x-1}$ is
a) [1,2] b) [−1,1]
c) [0,1] d) [−1,0]

10) If $\vec{a}\times(\vec{b}\times\vec{c}) = (\vec{a}\times\vec{b})\times\vec{c}$, where $\vec{a}, \vec{b}, \vec{c}$ are any three vectors such that $\vec{b}\cdot\vec{c} \neq 0$ and $\vec{a}\cdot\vec{b} \neq 0$, then $\vec{a}$ and $\vec{c}$ are
a) perpendicular b) parallel
c) inclined at an angle $\frac{\pi}{3}$ d) inclined at an angle $\frac{\pi}{6}$

11) The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is
a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com

www.Padasalai.Net www.TrbTnpsc.com

12) If $\rho(A) = \rho([A \mid B])$, then the system AX = B of linear equations is
a) consistent and has a unique solution
b) consistent
c) consistent and has infinitely many solution
d) inconsistent

13) If x + y = k is a normal to the parabola $y^2 = 12x$, then the value of k is
a) 3 b) −1 c) 1 d) 9

14) The equation of the circle passing through (1,5) and (4,1) and touching y-axis is $x^2 + y^2 - 5x - 6y + 9 + \lambda(4x + 3y - 19) = 0$ where λ is equal to
a) $0, \frac{-40}{9}$ b) 0 c) $\frac{40}{9}$ d) $\frac{-40}{9}$

15) A zero of $x^3 + 64$ is
a) 0 b) 4 c) 4i d) −4

16) If α, β and γ are the zeros of $x^3 + px^2 + qx + r$, then $\sum \frac{1}{\alpha}$ is
a) $-\frac{q}{r}$ b) $-\frac{p}{r}$ c) $\frac{q}{r}$ d) $-\frac{q}{p}$

17) If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$, then the values of x and y are respectively
a) $e^{(\Delta_2/\Delta_1)}, e^{(\Delta_3/\Delta_1)}$ b) $e^{(\Delta_1/\Delta_3)}, e^{(\Delta_2/\Delta_3)}$ c) $\log(\Delta_2/\Delta_1), \log(\Delta_3/\Delta_1)$ d) $\log(\Delta_1/\Delta_3), \log(\Delta_2/\Delta_3)$

18) The angle between the lines $\frac{x-2}{3} = \frac{y+1}{-2}$, z = 2 and $\frac{x-1}{1} = \frac{2y+3}{3} = \frac{z+5}{2}$ is
a) $\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $\frac{\pi}{3}$ d) $\frac{\pi}{2}$

19) If α and β are the roots of $x^2 + x + 1 = 0$, then $\alpha^{2020} + \beta^{2020}$ is
a) −2 b) −1 c) 1 d) 2

20) If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$, then
a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < c < 1$

Part-B

Answer any SEVEN questions: **7×2=14**
Question number 30 is compulsory.

21) Find the equation of the circle with centre (2,−1) and passing through the point (3,6) in standard form.

22) Find the angle between the line $\vec{r} = (2\hat{i} - \hat{j} + \hat{k}) + t(\hat{i} + 2\hat{j} - 2\hat{k})$ and the plane $\vec{r} \cdot (6\hat{i} + 3\hat{j} + 2\hat{k}) = 8$.

23) If $|z| = 3$, show that $7 \le |z + 6 - 8i| \le 13$.

24) Find the volume of the parallelepiped whose coterminous edges are represented by the vectors $-6\hat{i} + 14\hat{j} + 10\hat{k}$, $14\hat{i} - 10\hat{j} - 6\hat{k}$ and $2\hat{i} + 4\hat{j} - 2\hat{k}$.

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com

25) Which one of the points i, $-2 + i$ and 3 is farthest from the origin?

26) If p is real, discuss the nature of the roots of the equation $4x^2 + 4px + p + 2 = 0$ in terms of p.

27) Find adj(adj(A)) if $\text{adj}A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.

28) Find the value of $\tan^{-1}\left(\tan\frac{5\pi}{4}\right)$.

29) Solve the following system of linear equations, using matrix inversion method: $5x + 2y = 3$, $3x + 2y = 5$.

30) Find the equation of the parabola with focus $\left(-\sqrt{2}, 0\right)$ and directrix $x = \sqrt{2}$.

Part-C

Answer any SEVEN questions: **7×3=21**

Question number 40 is compulsory.

31) Find the sum of the squares of the roots of $ax^4 + bx^3 + cx^2 + dx + e = 0$, $a \neq 0$.

32) Simplify $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3$ into rectangular form.

33) Find the magnitude and direction cosines of the torque of a force represented by $3\hat{i} + 4\hat{j} - 5\hat{k}$ about the point with position vector $2\hat{i} - 3\hat{j} + 4\hat{k}$ acting through a point whose position vector is $4\hat{i} + 2\hat{j} - 3\hat{k}$.

34) If $A = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -3 \\ 5 & 2 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.

35) Find the domain of $\cos^{-1}\left(\frac{2+\sin x}{3}\right)$.

36) The equation $y = \frac{1}{32}x^2$ models cross sections of parabolic mirrors that are used for solar energy. There is a heating tube located at the focus of each parabola; how high is this tube located above the vertex of the parabola?

37) If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$.

38) If $\omega \neq 1$ is a cube root of unity, then show that $\frac{a + b\omega + c\omega^2}{b + c\omega + a\omega^2} + \frac{a + b\omega + c\omega^2}{c + a\omega + b\omega^2} = -1$.

39) Write in polar form the following complex number: $-1 - i$.

40) Find the image of the point whose position vector is $\hat{i} + 2\hat{j} + 3\hat{k}$ in the plane $\vec{r} \cdot (\hat{i} + 2\hat{j} + 4\hat{k}) = 38$.

www.Padasalai.Net

www.Padasalai.Net www.TrbTnpsc.com

Part-D

Answer all the questions: **7×5=35**

41) a) The prices of three commodities A, B and C are Rs.x, y and z per units respectively. A person P purchases 4 units of B and sells two units of A and 5 units of C. Person Q purchases 2 units of C and sells 3 units of A and one unit of B. Person R purchases one unit of A and sells 3 units of B and one unit of C. In the process, P, Q and R earn Rs.15,000, Rs.1,000 and Rs.4,000 respectively. Find the prices per unit of A, B and C. (Use matrix inversion method to solve the problem.) **(OR)**

b) Prove by vector method that $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$.

42) a) If $z = x + iy$ and $\arg\left(\frac{z-i}{z+2}\right) = \frac{\pi}{4}$, then show that $x^2 + y^2 + 3x - 3y + 2 = 0$.

(OR)

b) Solve the following system of linear equations by Gaussian elimination method: $4x + 3y + 6z = 25$, $x + 5y + 7z = 13$, $2x + 9y + z = 1$.

43) a) On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4 m when it is 6 m away from the point of projection. Finally it reaches the ground 12 m away from the starting point. Find the angle of projection. **(OR)**

b) A tunnel through a mountain for a four lane highway is to have an elliptical opening. The total width of the highway (not the opening) is to be 16 m, and the height at the edge of the road must be sufficient for a truck 4 m high to clear if the highest point of the opening is to be 5 m approximately. How wide must the opening be?

44) a) Find the parametric form of vector equation and Cartesian equations of the plane containing the line $\vec{r} = (\hat{i} - \hat{j} + 3\hat{k}) + t(2\hat{i} - \hat{j} + 4\hat{k})$ and perpendicular to the plane $\vec{r}.(\hat{i} + 2\hat{j} + \hat{k}) = 8$.

b) Prove by vector method that the perpendiculars (attitudes) from the vertices to the opposite sides of a triangle are concurrent.

45) a) Identify the type of conic and find centre, foci, vertices, and directrices of each of the following: $\frac{(y-2)^2}{25} - \frac{(x+1)^2}{16} = 1$. **(OR)**

b) Find all cube roots of $\sqrt{3} + i$.

46) a) Solve the following equation: $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$. **(OR)**

b) Find the value of $\cot^{-1}(1) + \sin^{-1}\left(\frac{-\sqrt{3}}{2}\right) - \sec^{-1}(-\sqrt{2})$.

47) a) Solve the equation $3x^3 - 26x^2 + 52x - 24 = 0$ if its roots form a geometric progression. **(OR)**

b) Solve $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \frac{\pi}{4}$.

- - - - -

Kindly Send Your Study Materials To Us Email ID: padasalai.net@gmail.com